

06/07/2026 - SNR & SRT old data recalibration

Issue:

To first understand the issue in the old data, let's understand the old code:

In the old code, we had:

```
targetNoiseVolume = targetNoiseVolume + store.rightOffset + store.listenLevel  
let targetDigitsVolume = 0 + store.leftOffset + store.listenLevel  
store.currentSNR = targetDigitsVolume - targetNoiseVolume;
```

The SNR was calculated using the right and left offset values baked into the formula, which let the starting SNR to be non-zero when right and left offsets are not the same. However, the targetNoiseVolume and targetDigitsVolume do map to the correct presentation value dB when played on the speakers - so that part of the experiment still holds true.

The fix we have in the new code is this:

```
store.currentSNR = -targetNoiseVolume;  
targetNoiseVolume = targetNoiseVolume + store.listenLevel -  
store.rightZeroSPL  
let targetDigitsVolume = store.listenLevel - store.leftZeroSPL
```

The SNR is not affected by the offset values, rather, it is just -targetNoiseVolume. The variable targetNoiseVolume is just the number used to adjust the noise volume after every patient number guess. The variable targetNoiseVolume starts at 0 (because at the start we don't adjust noise at all) and is decremented or incremented by 2. As such, we treat -targetNoiseVolume as the SNR because it essentially tracks the difference between noise and digit since we know that at the start of the code, they both differ by 0.

Now in the old data, we see incorrect starting snr values and, as such, incorrect srt values. SRT values are just the average of the snr's from runs 7 to 25. An example incorrect output would be this:

```
{"srt":5.526315789473684,
```

```
"responses":  
[{"response":"324","target":"324","snr":9,"correct":true},  
{"response":"193","target":"193","snr":7,"correct":true},  
{"response":"844","target":"780","snr":5,"correct":false},  
...
```

Here, the starting snr is 9, when it should be 0. However, the SNR values are incremented and decremented by 2, which is expected behaviour. The SNR values not starting with 0 can be seen because of the issues in the old code (see old code issue above).

The fix here is to simply: Offset every SNR value by the starting SNR we get in the old incorrect data. The old code still plays the correct presentation dB values for left and right channels, meaning that the starting SNR is 0. The fault is just in the calculation of the SNR. After adjusting all the SNR values, we then simply correct the SRT value by recalculating the average SNRs over runs 7 - 25.